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E. Nearly Shortest Repeating Substring

time limit per test: 2 seconds
memory limit per test: 256 megabytes
input: standard input
output: standard output

You are given a string s of length n consisting of lowercase Latin characters. Find the length of the shortest string k such that several (possibly one) copies of k can be concatenated together to form a string with the same length as s and, at most, one different character.

More formally, find the length of the shortest string k such that $c = \underbrace{k + \cdots + k}_{x \text{ times}}$ for some *positive integer* x, strings s and c has the same length and $c_i \neq s_i$ for at most one i (i.e. there exist 0 or 1 such positions).

Input

The first line contains a single integer t ($1 \leq t \leq 10^3$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of string s.

The second line of each test case contains the string s, consisting of lowercase Latin characters.

The sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print the length of the shortest string k satisfying the constraints in the statement.

Codeforces Round 937 (Div. 4)
Finished
Practice

→ Virtual participation

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Start virtual contest

→ Submit?

Language: Java 21 64bit
Choose file: Choose File No file chosen
Submit

Example

input Copy

```
5
4
abaa
4
abba
13
slavicgslavic
8
hshahaha
20
stormflamestornflame
```

output Copy

```
1
4
13
2
10
```

Note

In the first test case, you can select $k = \mathtt{a}$ and $k + k + k + k = \mathtt{aaaa}$, which only differs from s in the second position.

In the second test case, you cannot select k of length one or two. We can have $k = \mathtt{abba}$, which is equal to s.

→ Problem tags

implementation number theory strings

No tag edit access

→ Contest materials

- Announcement (en)
- Tutorial (en)

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